Maximum Blackouts That Keep Every Lattice Rectangle Identifiable

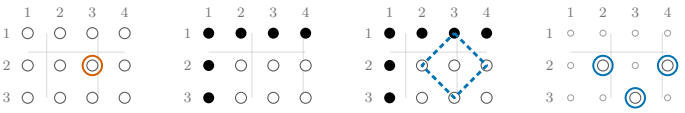
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Abstract

Kagey's Problem 001 concerns a grid of points, some of them blacked out. A rectangle with its corners on grid points, tilted or not, shows only its white corners. We ask how many points can be blacked out while every rectangle stays determined by what it shows. We study the largest such blackout and the number of blackouts of that size, the tables A397315 and A399580 of the On-Line Encyclopedia of Integer Sequences. A graph on the rows and columns and Mantel's theorem bound the maximum by the width plus a quarter of the squared height. We find the maximum on up to five rows at every width, and on six rows from width 15 on. On five rows from width 8 on, and on the 6×8 grid, the maximum exceeds the sum of the two sides, against a guess we recorded there. On three rows we count the optima exactly, and on four rows we find their growth rate. For every fixed number of rows the maximum minus the width is eventually periodic. Some values rest on computer checks.

1 Introduction

You are shown a grid of points with h rows and m columns, and you black out some of them. We call the set of blacked-out points the *blackout*. Then



(a) the point $(3, 2)$ is ringed (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on the 3×4 grid. Black dots are blacked out and open dots are white. In (d) the gray specks are points we are told nothing about.

the game master draws a rectangle with its four corners on grid points, tilted or not. You are told which of its corners are white, and you must say where all four corners are. Kagey asks how many points you can black out so that no two rectangles ever look the same.

We write (i, j) for the point in column i and row j, counting columns from the left and rows from the top. Figure 1 plays one round on the 3×4 grid with row 1 and column 1 blacked out. The game master draws the tilted square with corners $(2, 2)$, $(3, 1)$, $(4, 2)$, and $(3, 3)$, and we are shown $(2, 2)$, $(4, 2)$, and $(3, 3)$. Three corners of a rectangle determine the fourth (Lemma 4), so the fourth corner is $(3, 1)$. This blackout of six points is good, and no good blackout of the 3×4 grid is larger (Section 5).

Kagey poses the problem in his Open Problem Collection [12] and in his gift for the fourteenth Gathering 4 Gardner [11]. We write $f(h, m)$ for the largest size of a good blackout of the $h \times m$ grid, and $a_h(m)$ for the number of good blackouts of that size, counted with their coordinates. From Section 2 on, a good blackout is called *valid* (Definition 1). These are the tables A397315 and A399580 of the OEIS [18], which we contributed. Our comments there record the conjecture $f(3, m) = m + 2$ and the guess that $f(h, m) = h + m$ whenever $\min(h, m) \geq 4$, apart from $f(4, 4) = 7$ and $f(4, 5) = 8$. We show that the guess fails. The three-row conjecture was proved in Lean by The Omega Institute [19]. It is also the case $h = 3$ of our row bound, and we do not claim it as new.

Results. Two rectangles look the same exactly when every corner that belongs to only one of them is black (Lemma 2). Put $q_h = \lfloor h^2/4 \rfloor$. Computer checks are marked in each statement.

Theorem A (narrow grids)**.** *(a) For all $h, m \geq 1$ we have $f(h, m) \geq h + m - 1$, and for $m \geq 3$ we have $f(h, m) \leq m + q_h$.*

(b) *For $m \geq 3$ we have $f(2,m) = m+1$ and $f(3,m) = m+2$. Also $f(4,m) = m+4$ for every $m \geq 6$, and $f(5,5) = 10$.*
(c) *(Computer-assisted.) We have $f(4,4) = 7$, $f(4,5) = 8$, $f(5,6) = 11$, and $f(5,7) = 12$, while $f(5,m) = m+6$ for every $m \geq 8$.*
(d) *(Computer-assisted for $m \geq 9$.) For $8 \leq m \leq 14$ we have $f(6,m) \leq m+8$, and $f(6,m) = m+9$ for every $m \geq 15$.*
(e) *(Computer-assisted.) We have $f(6,6) = 12$, $f(6,7) = 13$, $f(7,7) = 14$, and $f(6,8) = 15$.*

The lower bound in (a) is the boundary cross of Lemma 16, and the upper bound is the row bound of Theorem 8, proved with Mantel's theorem. So for $h \leq 6$ the row bound is attained at every large width, and the guess $f(h,m) = h+m$ first fails on the 5×8 grid (Theorem 44).

Theorem B (three-row optima)**.** *For $m \geq 4$ we have $a_3(m) = 4(m-1)(3m+2)\,3^{m-5} + E_m$, where $E_m \geq 0$ counts the optima with a column black in rows 1 and 3. So $a_3(m) \sim \frac{4}{81} m^2 3^m$, and a fraction tending to $4/27$ of the $m^2 3^{m-1}$ axis-valid candidates is valid. The series $\sum_{m \geq 3} a_3(m) z^m$ is rational. (Computer-assisted.) Its reduced denominator is $(1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$, and E_m has a closed form with $E_m \sim \frac{19+11\sqrt{3}}{432} m^2 (1+\sqrt{3})^m$.*

These are Theorems 24, 27, and 25, with the last proof finished in Appendix A. The sequence starts $53, 224, 892, 3420, 12704$ at $m = 3$.

Theorem C (four rows)**.** *Let $m \geq 4$. No valid blackout of the $4 \times m$ grid hides all four corners of a rectangle. For $m \geq 6$,*

$$a_4(m) \leq 3\,(m)_4\,4^{m-4}\Big(\frac{53}{64}\Big)^{\max(0,\lfloor m/4 \rfloor - 4)}, \qquad (m)_4 = m(m-1)(m-2)(m-3).$$

(Computer-assisted.) The series $\sum_{m \geq 6} a_4(m) z^m$ is rational with reduced denominator of degree 420, and $a_4(m) \sim c\, m^4 \rho_4^m$ with $c = 8.1611 \times 10^{-5}$ and $\rho_4 = 3.35199\ldots$, an algebraic number of degree 20.

The bound has the base $212^{1/4} = 3.8157\ldots < 4$, so valid blackouts become rare (Theorems 32, 34, and 37).

Theorem D (finite collar)**.** *Let $h \geq 3$ and $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$. Pad a blackout of an $h \times m$ grid, on one side or on both, by columns black only in the top row, or only in the bottom row. Then every such padding is valid if and*

only if the padding by W_h columns on each chosen side is valid. (Computer-assisted.) A blackout of the 5×8 grid and one of the 6×16 grid pass this test with the top row on the right. Hence $f(5, m) = m + 6$ for $m \geq 8$ and $f(6, m) = m + 9$ for $m \geq 16$, and a stored blackout of the 6×15 grid gives $f(6, 15) = 24$.

The collar theorem is Theorem 39, and Section 7 has the two blackouts. The padding row must be a boundary row (Remark 40).

Theorem E (every fixed height)**.** *Fix $h \geq 1$ and write a blackout of the $h \times m$ grid as the word $B_1 \cdots B_m$, where B_i is the set of black rows of column i. The valid words form a regular language. If $h \geq 2$, the excess $f(h, m) - m$ is eventually periodic in m, and $\sum_{m \geq 0} a_h(m) z^m$ is rational.*

These are Theorems 48 and 49, and they give no period and no threshold.

Prior work. In its hitting-set form (Corollary 3) the game asks for a test cover [13, 3, 8]. The axis-aligned rectangles alone lead to bipartite graphs without cycles of length four and six (Lemma 7), studied by Erdős [5] and by de Caen and Székely [4]. Neuwirth [17, Theorem 1 and Proposition 17] proves our row bound for them when $m \geq q_h$. Our proof uses Mantel's theorem [14], covers every $m \geq 3$, and counts the equality case. None of these works sees tilted rectangles. Grids of fixed height are often treated by column transfer [6, 15, 1], [7, Proposition V.6]. Jiang [10] proves that optimal identifying codes in strips can be taken periodic, and Theorem E is of the same kind. It is close to a theorem of Guilhot, Little, and Parkinson [9, Theorem 1], but we maximize separately at each width and prove it directly. The code and data are in the repository [20].

2 Conventions and the difference lemma

These conventions hold throughout the paper.

(1) An $h \times m$ grid has h rows and m columns, and V is its set of points. A $2 \times m$ grid is a *strip*, and an $n \times n$ grid is a *square*.
(2) The point (i, j) is in column i and row j, and $(1, 1)$ is the top-left point. So i grows to the right and j grows downward, as in Figure 1(a).
(3) A *rectangle* is a set of four grid points that are the corners of a rectangle of positive area, and tilted rectangles count. We identify a rectangle R

with its corner set $C(R)$. It is *axis-aligned* if its sides are parallel to the rows and columns, and *tilted* otherwise. Four distinct points form a rectangle if and only if they split into two segments with the same midpoint and the same length.

(4) A *blackout* is a set $S \subseteq V$, and $K = V \setminus S$ is the *kept set*. The *presentation* of a rectangle R is the set $P_S(R) = C(R) \setminus S$ of its white corners.

(5) The *black rows* of column i form the set B_i of rows j with $(i, j) \in S$. Column i is *empty*, a *singleton*, a *double*, or a *triple* when $|B_i|$ is 0, 1, 2, or 3. It is *full* when $|B_i| = h$.

(6) The *pair set* F is the set of row pairs contained in some B_i. We put $q_h = \lfloor h^2/4 \rfloor$.

Definition 1 (valid blackout)**.** A blackout S is *valid* if different rectangles always have different presentations, that is, if $P_S(R_1) \neq P_S(R_2)$ whenever $R_1 \neq R_2$. The *maximum blackout* $f(h, m)$ is the largest size of a valid blackout of the $h \times m$ grid. We write $a_h(m)$ for the number of valid blackouts of size $f(h, m)$. They are counted with their coordinates, not up to symmetry.

Swapping rows and columns gives $f(h, m) = f(m, h)$ and $a_h(m) = a_m(h)$. The tables $f(h, m)$ and $a_h(m)$ are A397315 and A399580, read by antidiagonals with both indices starting at 2. Section 8 also uses width 0 and height 1, with no rectangles, where $f(h, 0) = 0$, $a_h(0) = 1$, $f(1, m) = m$, and $a_1(m) = 1$.

The empty presentation is allowed. It means that every corner is black, so a valid blackout hides at most one rectangle.

Lemma 2 (difference sets)**.** *For all sets A, B, and S we have $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$, where $\triangle$ is the symmetric difference.*

Proof. Both conditions say that no point outside S lies in exactly one of A and B. □

Corollary 3 (hitting sets)**.** *A blackout S is valid if and only if the kept set K meets $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles R_1, R_2. So $hm - f(h, m)$ is the least size of a set meeting every such difference set.*

Proof. Apply Lemma 2 to $C(R_1)$, $C(R_2)$, and S for every pair. □

A larger kept set meets more sets, so every subset of a valid blackout is valid. Hence a search that finds no valid blackout of exactly k points proves $f(h, m) < k$.

Lemma 4 (three corners suffice)**.** *Three corners of a rectangle determine the rectangle. So two distinct rectangles with the same presentation both show at most two corners.*

Proof. Three corners p, q, and r of a rectangle form a right triangle, with the right angle at the corner adjacent to both others, say q. A triangle has only one right angle, so the three points determine q and the fourth corner $p + r - q$. □

3 Strips

On two rows every rectangle is axis-aligned. Indeed, a tilted rectangle has no horizontal side, so both sides at a lowest corner go strictly up, and the opposite corner lies in a third row.

Theorem 5 (Strip Theorem)**.** *Let $m \geq 3$.*

(a) A blackout of the $2 \times m$ strip is valid if and only if at most one of its columns is full.

(b) We have $f(2, m) = m + 1$, and the maximum blackouts are those with one full column and one black point in every other column.

(c) There are $a_2(m) = m \cdot 2^{m-1}$ maximum blackouts.

For $m = 2$ there is only one rectangle, so every blackout is valid, $f(2, 2) = 4$, and $a_2(2) = 1$.

Proof. (a) Two distinct rectangles share at most one column. If they share one, it cancels, and the difference set is the four points of the other two columns p and q, the *pair constraint* on $\{p, q\}$. If they share no column, the difference set is the eight points of four columns, and it contains a pair constraint. Every pair constraint occurs, since $m \geq 3$ gives a third column z, and the rectangles on $\{z, p\}$ and $\{z, q\}$ have exactly this difference set. By Corollary 3, S is valid if and only if K meets every two columns, that is, if and only if at most one column has no kept point.

(b) By (a) a valid S has $|K| \geq m - 1$, so $|S| \leq m + 1$. Equality holds exactly when one column is full and every other column has one kept point, hence one black point. Such blackouts are valid by (a).

(c) There are m choices for the full column and two in each other column. □

So row 2 of A399580 is A001787$(m) = m2^{m-1}$ from $m = 3$ on.

Remark 6 (orbits). Up to the row reflection and the column reversal, there are $o_m = m\,2^{m-3}+[m \text{ odd}]\,2^{(m-3)/2}$ maximum blackouts of the $2\times m$ strip for $m \geq 3$, by Burnside's lemma (proof omitted). So $o_3, \ldots, o_8$ are $4, 8, 22, 48, 116, 256$. These six values agree with A322284$(m-1)$, which counts nonattacking kings on a $2 \times (2m-2)$ board up to symmetry.

4 The row-column graph and general bounds

This section proves $h + m - 1 \leq f(h, m) \leq m + q_h$ and lists the tilted rectangles of narrow grids. A blackout S of the $h \times m$ grid is *axis-valid* if no two axis-aligned rectangles have the same presentation. A rectangle that shows at most two corners is *weak*, and by Lemma 4 only weak rectangles can be confused.

The *row-column graph* G_S has a vertex r_j for each row j, a vertex c_i for each column i, and an edge $r_j c_i$ for each black point (i, j). A fully black axis-aligned rectangle on rows j, j' and columns i, i' is the cycle $r_j\, c_i\, r_{j'}\, c_{i'}$.

Lemma 7 (axis-aligned rectangles and short cycles)**.** *If G_S has no four-cycle and no six-cycle, then S is axis-valid. If $m \geq 3$ and S is axis-valid, then G_S has no four-cycle and no six-cycle.*

Proof. Let $m \geq 3$. A four-cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ and a third column w give the rectangles on rows $\{j, j'\}$ and columns $\{i, w\}$ or $\{i', w\}$, which both show the white points among (w, j) and (w, j'). For a six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ the different rectangles on rows $\{a, b\}$, columns $\{x, y\}$ and on rows $\{a, c\}$, columns $\{y, z\}$ are black except possibly at (y, a).

Conversely, let $R_1 \neq R_2$ be axis-aligned with the same presentation, so $C(R_1) \triangle C(R_2) \subseteq S$ by Lemma 2. Opposite corners determine an axis-aligned rectangle, so the common corners lie in one row or one column. Two common corners (x, a) and (y, a) leave rows $b \neq c$ and the black four-cycle $r_b\, c_x\, r_c\, c_y$, and a common column is similar. If (x, a) is the only common corner, with R_1 on rows $\{a, b\}$ and columns $\{x, y\}$ and R_2 on rows $\{a, c\}$ and columns $\{x, z\}$, then $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$ is a black six-cycle. With no common corner, R_1 is a black four-cycle. The all-black 2×2 grid shows that $m \geq 3$ is needed. □

Theorem 8 (row bound)**.** *Let $m \geq 3$. Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points, and at most $h + q_m$ points if also $h \geq 3$. Hence $f(h, m) \leq m + q_h$ for $m \geq 3$.*

For each nonempty column i choose a tree T_i with vertex set B_i, and let H be the graph on the rows whose edges are those of all the trees T_i.

Lemma 9 (row graph)**.** *Let $m \geq 3$ and let S be axis-valid, with arbitrary trees T_i. Then H has no repeated edge and no triangle, and $e(H) = |S| - t$, where t is the number of nonempty columns.*

Proof. By Lemma 7, G_S has no four-cycle and no six-cycle. If columns $i \neq i'$ both gave the edge $r_j r_{j'}$, then $r_j\, c_i\, r_{j'}\, c_{i'}$ would be a four-cycle. So $e(H) = \sum(|B_i| - 1) = |S| - t$. A triangle $r_a r_b r_c$ does not lie in one tree. If two of its edges come from column i and one from column i', then two of $a, b, c \in B_i$ lie in $B_{i'}$, a four-cycle. If its edges come from columns i, i', i'', then $r_a\, c_i\, r_b\, c_{i'}\, r_c\, c_{i''}$ is a six-cycle. □

Lemma 10 (Mantel [14])**.** *Let H be a simple triangle-free graph on h vertices, with degrees $d(\cdot)$. Let v have maximum degree Δ, let A be its set of neighbors, and let B be the set of the other $h - \Delta$ vertices, with $e(B)$ edges inside B. Then*

$$e(H) = \sum_{b \in B} d(b) - e(B) \leq (h - \Delta)\Delta - e(B) \leq q_h.$$

Proof. No two vertices of A are adjacent, so every edge has an end in B, and the degree sum over B counts the edges inside B twice. Every degree is at most Δ, $e(B) \geq 0$, and $(h - \Delta)\Delta$ is an integer at most $h^2/4$. □

Proof of Theorem 8. Lemmas 9 and 10 give $|S| = t + e(H) \leq m + q_h$. Only Lemma 7 uses a third column, so for $h \geq 3$ the exchange of rows and columns gives $|S| \leq h + q_m$. □

This removes the hypothesis $m \geq q_h$ of Neuwirth's graph form [17, Theorem 1(ii)]. His extremal graphs [17, Proposition 17] are the blackouts of the next theorem, which adds the converse and the count.

Theorem 11 (equality structure)**.** *Let $h \geq 4$ and $m \geq 3$, and let S be a blackout of the $h \times m$ grid with $m + q_h$ points. Then S is axis-valid if and only if*

(a) every column is a singleton or a double,
(b) exactly q_h columns are doubles, with pairwise distinct row pairs, and
(c) these row pairs are the edges of a complete bipartite graph on the rows whose parts differ in size by at most one.

There are $p_h\,(m)_{q_h}\,h^{m-q_h}$ *such blackouts, where* $(m)_q = m(m-1)\cdots(m-q+1)$ *and* p_h*, the number of such unordered partitions of the rows, is* $\frac{1}{2}\binom{h}{h/2}$ *for even* h *and* $\binom{h}{\lfloor h/2\rfloor}$ *for odd* h*.*

Proof. Step 1. Equality in Mantel's theorem. Let S be axis-valid and build H from arbitrary trees. Since $|S| = m + q_h$, Lemmas 9 and 10 force $t = m$ and equality throughout Lemma 10. So every column is nonempty, $e(B) = 0$, and every $b \in B$ has its $\Delta = |A|$ neighbors in A. Hence H is complete bipartite with parts A and B, whose sizes differ by at most one since $(h-\Delta)\Delta = q_h$.

Step 2. No triple. Suppose $|B_i| \geq 3$, so T_i has a vertex v with two neighbors u, w, both in the part opposite to v. Since $h \geq 4$ the part of v has a vertex $z \neq v$. Replacing the edge vw of T_i by uw gives a tree on B_i, and the new graph H' keeps uz and wz since $z \neq v$. The triangle uwz contradicts Lemma 9 and proves (a).

Step 3. The doubles. By (a) the edges of H are the row pairs of the $e(H) = q_h$ doubles. Lemma 9 gives (b) and Step 1 gives (c).

Step 4. The converse. Under (a) and (b) no two columns share two black rows, and a six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ would need doubles $\{a,b\}$, $\{b,c\}$, $\{a,c\}$, a triangle in the graph of (c). So Lemma 7 applies.

Step 5. The count. Choose a partition, an injective assignment of its q_h edges to columns, and a row for each other column. The doubles determine these choices. □

Corollary 12 (small widths)**.** *Let* $h \geq 4$ *and* $3 \leq m < q_h$*. Then* $f(h,m) \leq m + q_h - 1$*. In particular* $f(5,5) \leq 10$*.*

Proof. Use Theorem 8. A valid blackout of $m + q_h$ points is axis-valid, so by Theorem 11(b) it has $q_h \leq m$ doubles. Note that $q_5 = 6$. □

For tilted rectangles put $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$, so $W_3 = 2$, $W_4 = 3$, and $W_5 = 5$.

Lemma 13 (leftmost corner)**.** *A tilted rectangle* R *has exactly one corner of smallest column and exactly one of smallest row. If* $\ell = (x, r)$ *is the leftmost corner, the sides from* ℓ *are* $(a, -b)$ *and* (c, d) *with positive integers* a, b, c, d *and* $ac = bd$*, and the corners are*

$$(x, r), \qquad (x+a,\, r-b), \qquad (x+c,\, r+d), \qquad (x+a+c,\, r-b+d). \quad (1)$$

So R *occupies the rows* $r-b$ *to* $r+d$*, its topmost corner is* $(x+a, r-b)$*, and its column span is* $a+c$*.*

Proof. No side is vertical, so two corners in the smallest column x are opposite and the center is in column x. Then the other two corners, with columns at least x and mean x, also lie in column x, which is impossible. Rows are similar. So both sides from ℓ go strictly to the right, and since they are perpendicular, one goes up and one goes down, with $ac = bd$. □

Lemma 14 (span bound)**.** *Let $h \geq 3$. Every tilted rectangle of the $h \times m$ grid has span $a + c \leq W_h$, so it meets at most $W_h + 1$ columns. Grids with one or two rows have no tilted rectangles.*

Proof. Here $b + d \leq h - 1$, and $(a - 1)(c - 1) \geq 0$ gives $a + c \leq bd + 1 \leq W_h$, since bd is an integer at most $(b + d)^2/4$. With at most two rows, $b + d \leq 1$ is impossible. □

For $1 \leq i \leq m - 2$ and $1 \leq j \leq h - 2$ the *diamond* $D(i, j)$ is the square with corners $(i, j + 1)$, $(i + 1, j)$, $(i + 2, j + 1)$, and $(i + 1, j + 2)$. Four rows allow four more kinds, A_i, B_i, C_i, and E_i for $1 \leq i \leq m - 3$. Each has one corner in each of the columns i to $i + 3$. Read from left to right, the rows of these corners are 3412 for A_i, 3142 for B_i, 2413 for C_i, and 2143 for E_i.

Lemma 15 (tilted rectangles on narrow grids)**.** *The tilted rectangles of the $3 \times m$ grid are the diamonds $D(i, 1)$ with $1 \leq i \leq m - 2$. Those of the $4 \times m$ grid are the diamonds $D(i, 1)$ and $D(i, 2)$ with $1 \leq i \leq m - 2$ and the rectangles $\mathsf{A}_i, \mathsf{B}_i, \mathsf{C}_i, \mathsf{E}_i$ with $1 \leq i \leq m - 3$.*

Proof. Each listed set is a rectangle. On three rows $b = d = 1$, so $a = c = 1$ and ℓ is in row 2. On four rows either $b = d = 1$, which gives the diamonds, or $\{b, d\} = \{1, 2\}$, $\{a, c\} = \{1, 2\}$, and ℓ is in row $1 + b$. The sides $(2, -2), (1, 1)$ give A, then $(1, -2), (2, 1)$ give B, $(2, -1), (1, 2)$ give C, and $(1, -1), (2, 2)$ give E. □

Lemma 16 (boundary cross)**.** *Let $h, m \geq 1$. The first row together with the first column of the $h \times m$ grid is a valid blackout. Hence $f(h, m) \geq h + m - 1$.*

Proof. Let $h, m \geq 2$. An axis-aligned rectangle on columns $x_1 < x_2$ and rows $y_1 < y_2$ shows $X \times Y$, where $X = \{x_1, x_2\} \setminus \{1\}$ and $Y = \{y_1, y_2\} \setminus \{1\}$ are nonempty. Since $|X| = 1$ forces $x_1 = 1$ and $|Y| = 1$ forces $y_1 = 1$, the set $X \times Y$ determines the rectangle. By Lemma 13 a tilted rectangle has at most two black corners, ℓ and the topmost one. If both are black, then $\ell = (1, 1 + b)$ and it shows $p = (1 + c, 1 + b + d)$ and $q = (1 + a + c, 1 + d)$, which

determine a, b, c, d. Since p and q differ in both coordinates, no axis-aligned rectangle shows $\{p, q\}$. Lemma 4 covers the presentations with three or more points. □

Lemma 17 (axis-aligned rivals)**.** *Suppose that no axis-aligned rectangle is fully black, as holds when S is axis-valid and $m \geq 3$. Call an axis-aligned rectangle that shows exactly a set P of at most two white points a* rival *of P. Then the empty set has no rival, and P has a rival if and only if it is a singleton $\{(x, r)\}$ with $\{r, s\} \in F$ for some $s \in B_x$, a vertical pair $\{(x, r), (x, s)\}$ with $\{r, s\} \in F$, a horizontal pair $\{(x, r), (z, r)\}$ with $B_x \cap B_z \neq \emptyset$, or an oblique pair $\{(x, r), (z, s)\}$ with $x \neq z$, $r \neq s$, and $(x, s), (z, r) \in S$.*

Proof. A fully black axis-aligned rectangle is a four-cycle of G_S, so the clause after "as holds when" is Lemma 7. A rival of a nonempty P has rows $\{r, s\}$ and columns $\{x, z\}$ as in the statement, with its other corners black. This says $s \in B_x$ and $\{r, s\} \subseteq B_z$ for a singleton, $\{r, s\} \subseteq B_z$ for a vertical pair, $s \in B_x \cap B_z$ for a horizontal pair, and $(x, s), (z, r) \in S$ for an oblique pair. Conversely, as $r \notin B_x$, these conditions force $z \neq x$ and $s \neq r$, and the rectangle on rows $\{r, s\}$ and columns $\{x, z\}$ shows exactly P. □

5 Three rows

On three rows the row bound of Theorem 8 is attained, and we count the maximum blackouts as words. Throughout, $m \geq 3$, the rows are $1, 2, 3$ from top to bottom, and the tilted rectangles are the diamonds $D(i, 1) = \{(i, 2), (i + 1, 1), (i + 2, 2), (i + 1, 3)\}$ for $1 \leq i \leq m - 2$ (Lemma 15).

Theorem 18 (three rows)**.** *For every $m \geq 3$, the maximum blackout of the $3 \times m$ grid has $m + 2$ points, that is, $f(3, m) = m + 2$.*

Proof. The upper bound is the case $h = 3$ of Theorem 8, since $q_3 = 2$. The blackout of row 1 and column 1 has $m + 2$ points. It is the boundary cross of Lemma 16 for $h = 3$, so it is valid. □

From now on every blackout has $m + 2$ points. We name the columns $a = \{1\}$, $b = \{2\}$, $c = \{3\}$, $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$ and $X = \{1, 2, 3\}$, and call a and c *outer*. A blackout with no empty column is its *word* $B_1 B_2 \cdots B_m$. We name each pair of F (Section 2) by its double letter.

Proposition 19 (the two types)**.** *Let $m \geq 3$ and let S be a blackout of the $3 \times m$ grid with $m + 2$ points.*

(a) If S is valid, its word has one letter X and singletons elsewhere (type T), or two different doubles and singletons elsewhere (type P).

(b) Conversely, for every word of type T or P the graph G_S is a tree, so no two axis-aligned rectangles have the same presentation.

(c) There are exactly $m^2 3^{m-1}$ words of type T or P.

Proof. (a) By Lemma 9, $|S| = t + e(H)$ with $t \leq m$ nonempty columns and $e(H) \leq 2$ (Lemma 10). So every column is nonempty and $\sum_i (|B_i| - 1) = 2$. Two equal doubles form a four-cycle in G_S, which Lemma 7 forbids.

(b) The column of X, or the two doubles, connect the three rows, and every other column hangs on one row. So G_S is a tree, and Lemma 7 applies. (c) Types T and P give $m3^{m-1}$ and $\binom{m}{2} \cdot 6 \cdot 3^{m-2} = m(m-1)3^{m-1}$ words. □

We call the words of type T or P *candidates*. They fall into the triple class X and the *ordered classes* AC, CA, AB, BA, BC and CB, named by the doubles from left to right. The *class* $\{A, C\}$ is $AC \cup CA$, and likewise for $\{A, B\}$ and $\{B, C\}$. Let AC_m be the set of valid words of width m in AC, and likewise for the others.

Lemma 20 (diamond test)**.** *Let S be a candidate with pair set F. Then S is valid if and only if every diamond $D(i, 1)$ passes the test below, where $P = D(i, 1) \setminus S$ is its presentation.*

(a) If $|P| \geq 3$ or P is empty, the diamond passes.

(b) If $P = \{(i + 1, 1), (i + 1, 3)\}$, the diamond fails if and only if $B \in F$.

(c) If $P = \{(i, 2), (i + 2, 2)\}$, it fails if and only if $B_i \cap B_{i+2}$ meets $\{1, 3\}$.

(d) If P is an oblique pair $\{(x, r), (z, s)\}$, it fails if and only if $(x, s), (z, r) \in S$.

(e) If $P = \{(x, r)\}$, it fails if and only if $\{r, s\} \in F$ for some $s \in B_x$.

Proof. Axis-aligned rectangles are told apart (Proposition 19(b)), and three points fix a rectangle (Lemma 4). Tests (b) to (e) are Lemma 17 for a diamond, since in (b) a rival needs $B \in F$ and in (e) it needs $\{r, s\} \in F$. An empty diamond has no axis-aligned rival, since G_S has no four-cycle. Two diamonds $D(i, 1)$ and $D(j, 1)$ with $i < j$ share at most one point, so a common presentation has at most one point. Then columns $i + 1$ and $j + 1$ are both black in rows 1 and 3, which no candidate allows. □

A *window* is a string of three consecutive letters, at columns i, $i+1$ and $i+2$. It *passes* or *fails* with the diamond $D(i,1)$. We call $(i,2)$ and $(i+2,2)$ the *tips*. A tip is black exactly when its letter is b, A, C or X.

Lemma 21 (singleton dichotomy)**.** *Let S be a candidate and let stu be a window of three singletons.*

(a) If $B \notin F$, the window passes.

(b) If $B \in F$, the window fails if and only if it is bab, bbb or bcb.

Proof. If $s = u = b$, the diamond shows the vertical pair (for $t = b$) or one point of column $i+1$ (for t outer), and tests (b) and (e) fail it exactly when $B \in F$. With one black tip it shows three corners or an oblique pair whose test (d) needs $(i+1,2)$, which is white. With no black tip three corners show. □

Lemma 22 (six of nine in the class $\{A,C\}$)**.** *Let S be a candidate of class $\{A,C\}$, so $F = \{A,C\}$, and let s,t be singletons.*

(a) A window sAt fails if and only if st is bb, bc or cb.

(b) A window sCt fails if and only if st is bb, ba or ab.

(c) A window with one double, at a tip, passes.

Proof. (a) Here $(i+1,1)$ and $(i+1,2)$ are black and $(i+1,3)$ is white. For $s = t = b$ the diamond shows $(i+1,3)$ alone, and test (e) fails it through $2 \in B_{i+1}$ and $C \in F$. For $s = b \neq t$ it shows $(i+2,2)$ and $(i+1,3)$, and test (d) asks whether $(i+2,3)$ is black, that is, whether $t = c$. The case $t = b \neq s$ is the mirror image, and with no black tip three corners show. Part (b) follows by the reflection in the middle row, which swaps a,c and A,C.

(c) Let the window be Ast, so the tip $(i,2)$ is black. If $t = b$, the diamond shows the vertical pair or one point of column $i+1$, and both pass by tests (b) and (e), since $B \notin F$. If $t \neq b$, it shows three corners or an oblique pair whose test (d) needs $(i+1,2)$ black, and that point is white. Reflection and reversal give the other windows. □

Heuristically, each double of the class $\{A,C\}$ passes for six of its nine neighbor pairs, and this class holds about a third of the candidates. So about $\frac{1}{3} \cdot (\frac{6}{9})^2 = \frac{4}{27}$ of the candidates are valid (Corollary 26).

By Lemma 20, for $m \geq 3$ the set AC_m consists of the words $UAVCW$ of width m, with U, V and W strings of singletons, whose windows all pass the test with $F = \{A,C\}$. We use the same description for $m = 2$, put

$G_{AC}(z) = \sum_{m\ge 2} |AC_m| z^m$, and define G_{AB} and G_X in the same way from the words $UAVBW$ and UXW, with $F = \{A, B\}$ and $F = \{A, B, C\}$. Put $R = 1 - 3z$. The strings of $n \ge 1$ singletons with a prescribed letter next to a double have the generating function z/R.

Proposition 23 (the class $\{A, C\}$)**.** *The generating function of the ordered class AC is*

$$G_{AC}(z) = \frac{z^2 - 4z^3 + 10z^4 - 8z^5 - 3z^6}{(1-3z)^3}, \tag{2}$$

so $|AC_2| = 1$, $|AC_3| = 5$ and

$$|AC_m| = 2(m-1)(3m+2)3^{m-5} \qquad (m \ge 4). \tag{3}$$

Proof. Step 1. The windows with both doubles. In a window tAC the points $(i+1, 1)$ and $(i+2, 2)$ are black and $(i+1, 3)$ is white. For $t = b$ the diamond shows $(i + 1, 3)$ alone, and test (e) fails it. For $t \ne b$ it shows $(i, 2)$ and $(i + 1, 3)$, and test (d) fails it exactly when $t = c$. By the half turn ACt passes only for $t = c$, and AtC passes by tests (b) and (e).

Step 2. Adjacent doubles. In $UACW$ every window passes except possibly the two of Step 1 (Lemmas 21(a) and 22(c)). So U ends in a and W begins with c, if nonempty, and this case contributes $z^2(1 + z/R)^2$.

Step 3. One singleton between. In $UAtCW$ the window AtC passes. By Lemma 22(a), the last letter of U has $p_t = 3, 1, 2$ allowed values for $t = a, b, c$, and by (b) the first letter of W has $q_t = 2, 1, 3$. So this case contributes $z^3 \sum_t (1 + p_t z/R)(1 + q_t z/R) = z^3(3 + 12z/R + 13z^2/R^2)$, since $\sum p_t = \sum q_t = 6$ and $\sum p_t q_t = 13$.

Step 4. At least two singletons between. In $UAv_1 \cdots v_k CW$ with $k \ge 2$ only uAv_1 and $v_k Cw$ can fail. If U is empty, v_1 is free, and otherwise six of the nine pairs (u, v_1) pass. So each end contributes $3+6z/R$, the free middle letters give $1/R$, and this case contributes $z^4(3+6z/R)^2/R = 9z^4(1-z)^2/R^3$.

Step 5. The coefficients. The three cases sum to (2). The coefficient of z^m in $z^k R^{-3}$ is $\binom{m-k+2}{2} 3^{m-k}$ for $m \ge k$ and 0 otherwise. For $m \ge 4$ the five terms with $k = 2, \ldots, 6$ sum to the polynomial $2(m-1)(3m+2)3^{m-5}$, and at $m = 2, 3$ they give 1 and 5. □

Theorem 24 (the number of three-row optima)**.** *For $m \ge 3$ let E_m be the number of maximum blackouts of the $3 \times m$ grid whose word is outside AC and CA. Then for every $m \ge 4$,*

$$a_3(m) = 4(m-1)(3m+2)3^{m-5} + E_m. \tag{4}$$

Proof. Column reversal maps AC, AB and BC onto CA, BA and CB, and the reflection in the middle row maps AB onto CB. Both preserve optima, so

$$a_3(m) = 2|AC_m| + 4|AB_m| + |X_m| = 2|AC_m| + E_m \qquad (m \geq 3), \tag{5}$$

and (3) gives (4). □

A column black in rows 1 and 3 is B or X, so E_m is as in Theorem B. At $m = 3$, $a_3(3) = 53 = 2 \cdot 5 + 4 \cdot 7 + 15$ by Propositions 52 and 51.

5.1 The generating function

Put $L = 1 - 2z - 2z^2$, $N = 1 + 2z^2$ and $D = LN$.

Theorem 25 (three-row generating function, parts (b) and (c) computer-assisted)**.**

(a) The series $\sum_{m\geq 3} a_3(m)z^m$ equals $2G_{AC} + 4G_{AB} + G_X - z - 12z^2$. Here G_{AC} is (2), and G_{AB} and G_X are the rational functions of Propositions 52 and 51, with denominators dividing D^3 and D^2.

(b) The reduced denominator of the series is exactly $Q = R^3L^3N^3$.

(c) With

$$\begin{aligned} P(z) = z &- 3z^2 - 34z^3 + 228z^4 - 641z^5 + 1307z^6 - 2186z^7 \\ &+ 2056z^8 - 236z^9 - 1060z^{10} + 2584z^{11} - 4264z^{12} \\ &- 1440z^{13} + 704z^{14} - 1024z^{15} + 2080z^{16} + 2176z^{17} + 384z^{18}, \end{aligned}$$

we have

$$\sum_{m\geq 3} a_3(m)z^m = \frac{P(z)}{Q(z)} - z - 12z^2. \tag{6}$$

Proof. (a) Use (5). The coefficients 1 and 12 of z and z^2 count the word X, the six words sX and Xs, and the six words of two different doubles.

(b) This is Theorem 53 in Appendix A. The orders at the roots of L and N rest on the computer-assisted form (16) of G_{AB}.

(c) Clearing denominators in (a) gives P, exactly in SymPy (`three_row_closed_form.py`). The script `three_row_closed_form_check.py` checks (6) against an independent transfer matrix with 350 states, in exact integers through width 100. □

Corollary 26 (asymptotics)**.** *Let* $\rho = 1+\sqrt{3}$*. As* $m \to \infty$*,*

$$a_3(m) = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m + O(m^2\rho^m). \tag{7}$$

So the valid fraction $a_3(m)/(m^2 3^{m-1})$ *of the candidates is* $\frac{4}{27} - \frac{4}{81m} - \frac{8}{81m^2} + O\big((\rho/3)^m\big)$*.*

Proof. The main term of (4) is $4(3m^2-m-2)3^{m-5} = \frac{4}{81}m^2 3^m - \frac{4}{243}(m+2)3^m$. By Theorem 25(a) and (5), E_m is the coefficient of z^m in $4G_{AB} + G_X$, whose poles are roots of D of order at most three. The root of least modulus is $1/\rho$, so $E_m = O(m^2\rho^m)$. □

Theorem 27 (three-row closed form, computer-assisted exact algebra)**.** *Let* $\sigma = 1-\sqrt{3}$*. For every* $m \geq 3$*,*

$$E_m = p_+(m)\rho^m + p_-(m)\sigma^m + T_m, \tag{8}$$

where $432\,p_\pm(m) = (19 \pm 11\sqrt{3})m^2 + (61 \pm 41\sqrt{3})m + 104 \pm 28\sqrt{3}$ *and* $T_m = (-2)^{\lfloor m/2 \rfloor}\,\tau(m)$ *with* $\tau(m) = (2m^2+41m-104)/216$ *for even* m *and* $\tau(m) = (7m^2-32m-175)/432$ *for odd* m*. Hence* $E_m \sim \frac{19+11\sqrt{3}}{432}m^2\rho^m$*, and* $E_m > 0$ *for every* $m \geq 4$*.*

Proof. Let c_m be the coefficient of z^m in $4G_{AB} + G_X$, whose denominator divides L^3N^3. By Theorem 25(a), $E_m = c_m$ for $m \geq 3$. The roots $1/\rho$, $1/\sigma$ and $\pm 1/(i\sqrt{2})$ of L^3N^3 are triple, so beyond a polynomial part of degree one the c_m have the form (8), and `three_row_closed_form.py` finds the twelve rational coefficients. Both sides satisfy the order-12 recurrence of L^3N^3 from $m = 14$ on, so agreement at $m = 2, \ldots, 13$ proves equality for every $m \geq 2$. The asymptotic follows since $|\sigma| < 1$ and $|T_m| = O(m^2 2^{m/2})$. Finally $E_m > 0$, since the cross $Xaa\cdots a$ of Theorem 18 lies in the triple class. □

Remark 28 (optima up to symmetry, computer-assisted)*.* Up to reflection and reversal there are 15, 58, 227, 864, 3190 and 11502 optima at widths 3 to 8. For even $m \leq 14$ only the half turn fixes optima, $2m3^{m/2-2}$ of them, so there are $(a_3(m) + 2m3^{m/2-2})/4$ orbits (`three_row_orbits.py`, proof omitted).

6 Four rows

On four rows the row bound of Theorem 8 is $m+4$, attained for every $m \geq 6$. We write $(m)_k = m(m-1)\cdots(m-k+1)$ and use the tilted rectangles $D(i,1)$, $D(i,2)$, A_i, B_i, C_i, E_i of Lemma 15.

Theorem 29 (four rows)**.** *For every $m \geq 6$ we have $f(4, m) = m + 4$.*

The upper bound is Theorem 8 with $h = 4$. For the lower bound we take

$$S = \{(i, 1) : 2 \leq i \leq m - 1\} \cup \{(m - 1, 2), (m, 2), (1, 3), (2, 3), (1, 4), (m, 4)\}, \tag{9}$$

which has $m+4$ points. The double columns are column 1 (rows $3, 4$), column 2 (rows $1, 3$), column $m - 1$ (rows $1, 2$), and column m (rows $2, 4$). So G_S is the eight-cycle $r_1\, c_2\, r_3\, c_1\, r_4\, c_m\, r_2\, c_{m-1}$ with the columns $c_3, \ldots, c_{m-2}$ hanging off r_1.

Proof of Theorem 29. Step 1. Axis-aligned rectangles. The graph G_S has no cycle of length four or six. So by Lemma 7 the axis-aligned rectangles have distinct presentations, and none is hidden.

Step 2. The weak tilted rectangles. By Lemma 15, a corner of a tilted rectangle in row 1 is always black, and one in row 4 is always white. The other black corners are $(i + 2, 2)$ for $i \geq m - 3$ and $(i + 1, 3)$ for $i = 1$ in $D(i, 1)$, and $(i, 3)$ for $i \leq 2$ and $(i + 1, 2)$ for $i = m - 2$ in $D(i, 2)$. In A_i and B_i they are $(i, 3)$ for $i \leq 2$ and $(i + 3, 2)$ for $i \geq m - 4$, and in C_i and E_i there are none. As $m \geq 6$, the two conditions for $D(i, 2)$ exclude each other. A rectangle with at most one black corner is safe by Lemma 4. So only the rectangles below need a check, with $s = 1$ for A_i and $s = 2$ for B_i.

rectangle	which i	corners shown
$D(1, 1)$		$(1, 2), (3, 2)$
$D(i, 1)$	$i \in \{m - 3, m - 2\}$	$(i, 2), (i + 1, 3)$
A_i or B_i	$i \in \{1, 2\}$, $i \notin \{m - 4, m - 3\}$	$(i + s, 4), (i + 3, 2)$
A_i or B_i	$i \in \{m - 4, m - 3\}$, $i \notin \{1, 2\}$	$(i, 3), (i + s, 4)$
A_i or B_i	$i = 2$, $m = 6$	$(i + s, 4)$

Step 3. Against each other. The rows of the shown points are $2, 2$ and $2, 3$ and $4, 2$ and $3, 4$ in the first four lines, so different lines never agree. Within a line the point in row 2 or 3 determines i, and the point in row 4 then determines s.

Step 4. Against axis-aligned rectangles. A rival of $\{(1, 2), (3, 2)\}$ needs a row r with $(1, r)$ and $(3, r)$ black, but there is none. In the next three lines the axis-aligned rectangle through the shown points has the white corner $(i, 3)$, $(i + s, 2)$, or $(i, 4)$, respectively, since $i \geq 3$, $i + s \leq 4 < m - 1$, and $3 \leq i \leq m - 3$ there. In the last line $m = 6$, and the shown point is $(z, 4)$

with $z = i + s \in \{3, 4\}$. Column z is black only in row 1, so a rival needs a column black in rows 1 and 4, and there is none. By Steps 1 and 2 no rectangle is hidden, so S is valid. □

Lemma 30 (equality in the row bound on four rows)**.** *Let $m \geq 4$. A blackout of the $4 \times m$ grid with $m+4$ points is axis-valid if and only if it has the form described next. Exactly four columns have two black points, their row pairs are the edges of a four-cycle on the rows, and every other column has one black point. There are $3\,(m)_4\,4^{m-4}$ such blackouts, which is 72 and 1,440 for $m = 4$ and 5.*

Proof. This is Theorem 11 with $h = 4$. There are three four-cycles on the rows, $(m)_4$ ways to place their labeled edges, and four rows for each other column. □

Corollary 31 (widths four and five, computer-assisted)**.** *We have $f(4, 4) = 7$ and $f(4, 5) = 8$.*

Proof. The lower bounds are Lemma 16. For the upper bounds, the script `verify_narrow.py` finds a repeated presentation in each of the 72 and 1,440 candidates of Lemma 30. The brute-force check of Section 9 confirms both values without the lemma. □

Theorem 32 (no hidden rectangle)**.** *Let $m \geq 4$. A valid blackout of the $4 \times m$ grid hides no rectangle.*

Proof. By Lemma 2, two rectangles have the same presentation exactly when the symmetric difference of their corner sets is black. So it suffices to write each corner set as $C(R_1) \triangle C(R_2)$ with $R_1 \neq R_2$. For the axis-aligned rectangle on rows r, s and columns x, y, take rows r, s with columns $\{x, w\}$ and $\{y, w\}$ for a third column w. For $1 \leq i \leq m-3$ let $\mathsf{Q}_i = \{(i, 2), (i, 3), (i+3, 2), (i+3, 3)\}$. Checking corners gives

$$\begin{aligned} \mathsf{C}_i &= \mathsf{A}_i \triangle \mathsf{Q}_i, & \mathsf{E}_i &= \mathsf{B}_i \triangle \mathsf{Q}_i, \\ D(i+1, 1) &= D(i, 2) \triangle \mathsf{A}_i, & D(i+1, 2) &= D(i, 1) \triangle \mathsf{E}_i, \end{aligned} \tag{10}$$

and each identity can be solved for each of its three rectangles. The first line handles A_i, B_i, C_i, and E_i. A diamond $D(j, 1)$ or $D(j, 2)$ has $j \geq 2$ or $j \leq m-3$, since $m \geq 4$, and the second line with $i = j - 1$ or $i = j$ handles it. □

The hypothesis $m \geq 4$ is needed, since the valid blackout $\{(1,2),(2,1),(3,2),(2,3)\}$ of the 4×3 grid hides $D(1,1)$ (checked by computer).

Let $m \geq 6$. By Lemma 30 an optimum has four double columns, whose pairs form a four-cycle F on the rows, and singletons elsewhere. We name F by its parts, $\{1,2\} \mid \{3,4\}$, $\{1,3\} \mid \{2,4\}$, or $\{1,4\} \mid \{2,3\}$. A *block* is a run of four consecutive singleton columns, and its *word* lists their black rows. The *template words* 2413, 3412, 2143, and 3142 are the corner rows of C_x, A_x, E_x, and B_x. A *swap word* exchanges two letters of a template word, for example 4213. An *edit word for F* replaces one letter r of a template word by a row s with $\{r,s\} \in F$.

Lemma 33 (forbidden blocks)**.** *Let S be a valid blackout of the $4 \times m$ grid with $m+4$ points and pair set F, where $m \geq 6$. No block of S has a swap word or an edit word for F. There are 20 swap words, and 24, 24, or 32 edit words for the three pair sets, none of them a swap word. So at most 212 of the 256 words of a block are allowed.*

Proof. Let the block start in column x, and let T be one of C_x, A_x, E_x, B_x, with template word $t_1t_2t_3t_4$. In both steps an axis-aligned rectangle has the same presentation as T. *Step 1. Swap words.* If the letters in positions $p \neq q$ are exchanged, T shows exactly $(x+p-1,t_p)$ and $(x+q-1,t_q)$, and so does the axis-aligned rectangle on these two points. The template words are the permutations with both end letters in $\{2,3\}$. Exchanging two end letters, or two middle letters, gives a template word again, and exchanging an end letter with a middle letter gives every permutation with exactly one end letter in $\{2,3\}$. These are the 20 swap words.

Step 2. Edit words. If $t_p = r$ is replaced by s with $\{r,s\} \in F$, then T shows only $(x+p-1,r)$. So does the rectangle on rows r,s and the columns $x+p-1$ and d, where d is the double column black in rows r,s. Each template word has 8 edit words.

Two template words share an edit word only if they differ in two positions, carrying letters r,s in opposite orders, and then they share two if $\{r,s\} \in F$. This happens for $2413, 3412$ and $2143, 3142$ with $\{r,s\} = \{2,3\}$, and for $2413, 2143$ and $3412, 3142$ with $\{r,s\} = \{1,4\}$. So there are $32 - 4[\{2,3\} \in F] - 4[\{1,4\} \in F]$ edit words. They repeat a letter, so they are not swap words. □

Theorem 34 (forbidden-block bound)**.** *For every $m \geq 6$,*

$$a_4(m) \leq 3\,(m)_4\,4^{m-4}\,(53/64)^k = O\big(m^4\,212^{m/4}\big), \tag{11}$$

where $k = \max(0, \lfloor m/4 \rfloor - 4)$.

Proof. Fix F, the double columns, and their pairs, in one of $3\,(m)_4$ ways. Of the $\lfloor m/4 \rfloor$ groups of columns $4k-3, \ldots, 4k$, at most four contain a double column. So at least k are blocks, each with at most $212 = 4^4 \cdot 53/64$ words by Lemma 33. Every other column has at most 4 values, which gives (11). □

So the valid fraction of the candidates tends to zero (compare Corollary 26).

Remark 35 (six columns of memory, proof omitted). Fix a pair set F. A finite automaton reads a blackout column by column, remembers its last six columns and the pairs of F already used, and accepts exactly the optima with pair set F. So their generating function is rational (Flajolet and Sedgewick [7, Section V.5]). The proof uses Lemma 17 and Theorem 32, and we omit it (see the repository notes [20]).

Proposition 36 (small widths, computer-assisted)**.** *In the row* $h = 4$ *of* A399580*,* $a_4(4) = 316$*,* $a_4(5) = 1{,}810$*,* $a_4(6) = 26$*, and* $a_4(7) = 292$*.*

Proof. The ILP program and the brute-force check of Section 9 both find these values. For $m = 6, 7$ the script `four_row_direct.py` tests the candidates of Lemma 30, and the automaton of Remark 35 in `four_row_counts.py` also gives 26 and 292 (Table 1). □

m	6	7	8	9	10	11	12
$a_4(m)$	26	292	2,118	13,332	75,398	402,656	2,037,848
$\{1,2\} \mid \{3,4\}$	0	4	128	920	4,852	24,112	114,424
$\{1,3\} \mid \{2,4\}$	20	256	1,780	11,128	63,540	346,320	1,785,768
$\{1,4\} \mid \{2,3\}$	6	32	210	1,284	7,006	32,224	137,656

Table 1: Four-row optimum counts by pair set, computer-assisted. Two transfer counters agree to width 1500 (data file `four_row_counts.json`), and a direct test of every candidate agrees to width 10.

Theorem 37 (the four-row generating function, computer-assisted)**.** *There are coprime integer polynomials* $P_4 = 26z^6 - 826z^7 + \cdots$ *of degree* 444 *and* $Q_4 = \sum_{j=0}^{420} q_j z^j$ *of degree* 420 *with* $q_0 = 1$ *and* $\sum_{m \geq 6} a_4(m) z^m = P_4/Q_4$*. The*

recurrence $\sum_j q_j\, a_4(n-j) = 0$ *holds for every* $n \geq 445$ *and fails at* $n = 444$. *Over the integers*

$$Q_4 = (1-z)^2(1+z)^2\, g_5^3\, \tilde{g}_5^3\, g_6^3\, g_{11}^5\, g_{17}^3\, g_{19}^3\, g_{20}^5\, g_{21}^5,$$

where g_d *and* $\tilde{g}_d$ *are irreducible of degree* d *with constant term* 1, *and all of* g_{20} *comes from the pair set* $\{1,3\} \mid \{2,4\}$.

Proof. The Berlekamp–Massey algorithm modulo eight primes, run on the values $a_4(m)$ for $m \leq 1500$ from the automaton of Proposition 36, gives P_4 and Q_4. With a register of used pairs the automaton has a transfer matrix T_F with 6,128, 7,792, or 4,912 states. Its state vectors $v(n) = v(0)T_F^n$ satisfy $\sum_j q_j v(n_F - j) = 0$ at $n_F = 436$, 435, and 445. Multiplying by powers of T_F gives the recurrence for every $n \geq 445$, and $\deg P_4 = 444$ forces the failure at $n = 444$. We factor Q_4 in SymPy, and `four_row_recurrence.json` lists the factors. □

Corollary 38 (growth, computer-assisted)**.** *Let* $\rho_4 = 3.351996777310\ldots$ *be the largest real root of the irreducible polynomial* $p(x) = x^{20}g_{20}(1/x) = x^{20} - 4x^{19}+2x^{18}+\cdots-26x^2-4x+8$. *Then* $a_4(m) \sim c\, m^4 \rho_4^m$ *with* $c = 8.1611\times10^{-5}$.

Proof. The script `four_row_root_isolation.py` isolates the complex roots of every factor of Q_4 in rational rectangles, in exact arithmetic in SymPy. The only root of modulus below 0.341 is the positive root $1/\rho_4 = 0.29832\ldots$ of g_{20}, which is simple in g_{20} and so has order five in Q_4. It is not a root of P_4, since the two are coprime. So c is the coefficient of $(1-\rho_4 z)^{-5}$ in the partial fraction divided by 4! (Flajolet and Sedgewick [7, Section IV.5]), computed to high precision. □

7 Five and six rows

One finite padding test settles every large width on five and on six rows.

7.1 A finite test for boundary padding

For a blackout S of the $h \times m$ grid and $L, R \geq 0$, let $S_{L,R}$ add L columns on the left and R on the right. They are the columns $1-L, \ldots, 0$ and $m+1, \ldots, m+R$. Each is black exactly in the *padding row*, which is row 1 or row h. As before, W_h is the span bound of Lemma 14.

Theorem 39 (finite collar)**.** *Let $h \geq 3$ and $m \geq 1$, and let S be a blackout of the $h \times m$ grid. Fix the padding row and choose the left side, the right side, or both. Call* (L, R) admissible *if $L = 0$ when the left side is not chosen and $R = 0$ when the right side is not chosen. Put $L^* = W_h$ if the left side is chosen and $L^* = 0$ otherwise, and define R^* in the same way. Then $S_{L,R}$ is valid for every admissible (L, R) if and only if S_{L^*,R^*} is valid.*

Proof. Write $S^* = S_{L^*,R^*}$. If every admissible padding is valid, then so is S^*. Conversely, let S^* be valid and $U = S_{L,R}$ with (L, R) admissible. By Lemma 13 the corners of a tilted rectangle lie in the rows j, $j-b$, $j+d$, $j+d-b$ with $b, d \geq 1$, so at most one lies in row 1 and at most one in row h. Hence a weak tilted rectangle of U has a black corner in an original column e, and by Lemma 14 its columns lie in $[e-W_h, e+W_h] \cap [1-L, m+R] \subseteq [1-L^*, m+R^*]$. Since U and S^* agree on their common columns, it is a rectangle of S^* with the same presentation. Suppose that two distinct rectangles R_1 and R_2 of U have the same presentation P.

Step 1. Both axis-aligned. The graph G_U is G_S plus one vertex of degree one for each added column, so its cycles are cycles of G_{S^*}. Since S^* is valid with at least three columns, neither graph has a cycle of length four or six, and U is axis-valid (Lemma 7).

Step 2. Both tilted. By Lemma 4, $|P| \leq 2$, so both are weak and are rectangles of S^* with the same presentation, which is impossible.

Step 3. One of each. Let R_1 be tilted, so it is a rectangle of S^*. A column of R_2 with a point of P is a column of R_1, and one without has two black corners and is an original column. So R_2 is a rectangle of S^* too, a contradiction. □

Remark 40 (the padding row has to be a boundary row). Take $h = 3$, $B_1 = B_2 = \{1\}$, $B_3 = \{1, 3\}$, and the interior padding row 2. The added columns 0 and -1 keep the board valid, but -2 breaks it. The diamond $(-2, 2)$, $(-1, 1)$, $(0, 2)$, $(-1, 3)$ and the rectangle on columns -1, 3 and rows 1, 3 both show $\{(-1, 1), (-1, 3)\}$.

7.2 Five rows

Let S_8 be the blackout of the 5×8 grid with $B_1 = \{1, 4\}$, $B_2 = \{4, 5\}$, $B_3 = \{2, 5\}$, $B_4 = B_5 = \{5\}$, $B_6 = \{3, 4\}$, $B_7 = \{2, 3\}$, and $B_8 = \{1, 2\}$. Its doubles carry the edges of $K_{2,3}$ between $\{2, 4\}$ and $\{1, 3, 5\}$. With five more

columns black in row 1 it gives the 5×13 board below, rows 1 to 5, with # for a black point.

```
#.....######  ..#...##.....  .....##......  ##...#.......  .####........
```

Theorem 41 (five rows, computer-assisted)**.** *(a) We have $f(5, m) = m + 5$ for $m = 5$, 6, and 7.*

(b) We have $f(5, m) = m + 6$ for every $m \geq 8$, attained by S_8 followed by $m - 8$ columns black in row 1 only.

(c) The 5×8 grid has exactly four maximum blackouts, the images of S_8 under the row reflection and the column reversal.

Proof. (b) The upper bound is Theorem 8. Apply Theorem 39 to S_8 with the padding row 1 on the right. Since $W_5 = 5$, the test board is the 5×13 board shown above. Three rectangle enumerators list its 954 rectangles and find no repeated presentation.

(a) The lower bounds are S_0 of Theorem 46 below for $m = 5$ and the stored blackouts of $h + m$ points (Section 9) for $m = 6, 7$. For $m = 5$ the upper bound is Corollary 12. For $m = 6$ and $m = 7$ a valid blackout of $m + 6$ points would be one of the $10\,(m)_6\,5^{m-6}$ candidates of Theorem 11. Two SAT encodings exclude all of them. Two more encodings and a column search, which use no structure theorem, agree.

(c) Among the $10\,(8)_6\,5^2 = 5{,}040{,}000$ candidates of Theorem 11, two SAT enumerations and a column search find exactly the four images of S_8. □

Remark 42 (five-row counts, computer-assisted). Table 2 gives $a_5(m)$ for $8 \leq m \leq 12$. For $m = 13, 14, \ldots, 23$ the numbers $a_5(m)$ are 193,406, 1,346,758, 8,650,416, 52,128,242, 299,349,210, 1,653,637,492, 8,828,301,226, 45,743,165,894, 230,934,909,328, 1,139,486,126,866, and 5,510,071,794,736. The values for $m \leq 12$ come from two independent programs and the others from two exact transfer counters, all based on Theorem 11. No proof uses them.

7.3 Six rows and small grids

Let S_{16} be the blackout of the 6×16 grid whose rows 1 to 6 are shown below.

```
.#...#.........#  ..#...........##  #...........##.
.#.......#####..  ..#.....##......  #..######.......
```

Its doubles, in the columns $1, 2, 3, 6, 9, 10, 14, 15, 16$, carry the edges of the complete bipartite graph between $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

Theorem 43 (six rows, computer-assisted)**.** *(a) For $8 \le m \le 14$ we have $f(6, m) \le m + 8$.*

(b) For every $m \ge 15$ we have $f(6, m) = m + 9$.

So for $m \ge 8$ we have $f(6, m) = m + 9$ if and only if $m \ge 15$.

Proof. (a) For $m = 8$ this is Corollary 12. For $9 \le m \le 14$, by Theorem 11 the doubles of a valid blackout of $m + 9$ points carry the edges of $K_{3,3}$ for one of ten row partitions. Two SAT encoders find none for any width and partition. A column search for $m \le 13$ and a third encoder for $m = 13$ and $m = 14$ give a third check at every width.

(b) The upper bound is Theorem 8. A valid blackout of 24 points on the 6×15 grid is in `jis/data/six_row_sat_w15_A.json`. For $m \ge 16$ apply Theorem 39 to S_{16} with the padding row 1 on the right. Since $W_6 = 7$, the test board adds $(17, 1), \ldots, (23, 1)$, and the three enumerators find no repeated presentation among its 4,627 rectangles. □

Theorem 44 (the guess fails, computer-assisted)**.** *(a) We have $f(6, 6) = 12$, $f(6, 7) = 13$, $f(7, 7) = 14$, and $f(6, 8) = 15$. The blackout $(1, 1)$, $(1, 5)$, $(2, 1)$, $(3, 1)$, $(3, 2)$, $(3, 3)$, $(4, 3)$, $(4, 4)$, $(5, 4)$, $(6, 4)$, $(7, 4)$, $(7, 5)$, $(7, 6)$, $(8, 2)$, $(8, 6)$ of the 6×8 grid attains the last value.*

(b) The guess of Section 1 fails. Indeed $f(5, 8) = 14$ and $f(6, 8) = 15$.

Proof. (a) The other lower bounds are stored blackouts of $h + m$ points. Encoding A of Section 9 on CaDiCaL and encoding B on Glucose and MiniSat find no valid blackout of 13, 14, 15, or 16 points on the 6×6, 6×7, 7×7, or 6×8 grid, respectively. A third encoding agrees on the last three, and the ILP program on the first.

(b) This is Theorem 41(b) and part (a). □

If a side is at most four, then $f(h, m) \le h + m$ by Sections 3 to 6. Also $f(h, m) = h + m$ on 5×5, 5×6, 5×7, and 6×6, the other grids of area below 40. So f first exceeds $h + m$ on the 5×8 grid.

Neuwirth [17, Remark 18] states the case $5+5$ of the next lemma without proof.

Lemma 45 (few edges without short cycles)**.** *Let G be a finite simple bipartite graph with v vertices, $1 \le v \le 10$, and with no cycle of length four or six.*

Then G has at most v edges. If it has exactly v edges, then G is connected and has exactly one cycle, of length 8 or 10.

Proof. Every cycle of G has length at least 8. Suppose a component has more edges than vertices. Removing vertices of degree one with their edges until none is left gives a connected H with more edges than vertices and minimum degree two, so some vertex x of H has degree at least three. Let D_k be the set of vertices at distance k from x. Two paths of length at most 3 from x to one vertex would close a cycle of length at most 6. So for $k = 1, 2, 3$ every vertex of D_k has exactly one neighbor in D_{k-1} and none in D_k. Its degree is at least two, so it also has a neighbor in D_{k+1}. For $k \leq 2$ no other vertex of D_k shares that neighbor. Hence $|D_1| \geq 3$, $|D_3| \geq |D_2| \geq |D_1|$ and $|D_4| \geq 1$, at least 11 vertices in all. So G has at most v edges, and with exactly v edges every component has one cycle. Two disjoint cycles of length at least 8 need 16 vertices, so G is connected. □

Theorem 46 (the 5×5 grid)**.** *(a) We have $f(5, 5) = 10$, and G_S is connected with exactly one cycle for every optimum S.*

(b) (Computer-assisted.) There are exactly $a_5(5) = 8$ maximum blackouts, the eight distinct images of S_0 below under the symmetries of the square.

Proof. Upper bound. If S is valid, G_S has 10 vertices, $|S|$ edges and no cycle of length four or six (Lemma 7), so Lemma 45 applies.

Lower bound. Let S_0 be the staircase of nine points plus the corner $(5, 5)$ with $B_1 = \{4, 5\}$, $B_2 = \{3, 4\}$, $B_3 = \{1, 2, 3\}$, $B_4 = \{1\}$, and $B_5 = \{1, 5\}$. Its graph is an 8-cycle with two pendant edges, so S_0 is axis-valid. By Lemma 13 the 30 tilted rectangles among the 130 are the nine unit diamonds, the four kinds of Lemma 15 in four positions each, one large diamond, and four that fill the grid. Exactly fifteen are weak. The table lists the two corners each one shows and, after a bar for an oblique pair, the white corner of its axis-aligned rectangle.

$(1,2),(2,1) \mid (1,1)$	$(1,3),(2,2) \mid (1,2)$	$(3,4),(4,3) \mid (4,4)$	$(2,5),(3,4) \mid (3,5)$
$(2,2),(4,2)$	$(1,2),(4,3) \mid (1,3)$	$(4,4),(5,2) \mid (4,2)$	$(3,5),(4,4) \mid (3,4)$
$(4,3),(5,2) \mid (4,2)$	$(3,4),(5,2) \mid (5,4)$	$(2,5),(5,2) \mid (2,2)$	$(4,5),(5,3) \mid (4,3)$
$(1,3),(4,2) \mid (1,2)$	$(2,5),(4,3) \mid (4,5)$		$(4,5),(5,4) \mid (4,4)$

The fifteen pairs are distinct, and the other tilted rectangles are determined by Lemma 4. Each listed white corner lies on the only axis-aligned rectangle through its pair, and $(2, 2), (4, 2)$ has no rival since $B_2 \cap B_4 = \emptyset$. So S_0 is valid.

(b). Of the 44,640 sets of 10 points without a short cycle, the script `verify_narrow.py` finds exactly the 8 images of S_0 valid. The ILP program and the brute-force check agree. □

8 Every fixed height

Fix h. A blackout of the $h \times m$ grid is a *column word* $w = B_1 B_2 \cdots B_m$ over the subsets of $\{1, \ldots, h\}$, with length $|w|$ and blackout $S(w)$. Let $\mathcal{L}_h$ be the set of words with a valid blackout, the empty word included. A *factor* of w is a word $B_i B_{i+1} \cdots B_j$.

Proposition 47 (validity is not a window property)**.** *Let $g \geq 1$. The blackout $S = \{(1,2), (3,2), (g+3,1), (g+3,3)\}$ of the $3 \times (g+3)$ grid is not valid, but every factor of at most g columns is valid.*

Proof. The diamond $D(1,1)$ and the rectangle on columns $2, g+3$ and rows $1, 3$ both show $\{(2,1), (2,3)\}$, while a factor of at most g columns has at most two black points and two different rectangles differ in at least four corners (Lemmas 2 and 4). □

Theorem 48 (regularity and rational series)**.** *For every $h \geq 1$ the language $\mathcal{L}_h$ is regular, and $G_h(z,u) = \sum_{w \in \mathcal{L}_h} z^{|w|} u^{|S(w)|}$ is rational in z and u.*

Proof. The *trace* of a rectangle R is the word whose ith letter is the set of rows of its corners in column i, and it determines $C(R)$. By Lemma 2, $w \notin \mathcal{L}_h$ exactly when two different traces Y, Y' of length $|w|$ have $Y_i \triangle Y'_i \subseteq B_i$ for every i. The traces form a regular language, $\emptyset^*\{r,s\}\emptyset^*\{r,s\}\emptyset^*$ for axis-aligned rectangles and, by Lemmas 13 and 14, $\emptyset^* Y_0 Y_1 \cdots Y_n \emptyset^*$ for finitely many templates (1) of span $n \leq W_h$. So a finite automaton that reads w and two traces in parallel recognizes the complement of $\mathcal{L}_h$, and $\mathcal{L}_h$ is regular. With the weight $u^{|B|}$ on a letter B, the transfer matrix $M(u)$ of a deterministic automaton for $\mathcal{L}_h$ gives $G_h = \alpha(I - zM(u))^{-1}\beta$, rational by Cramer's rule [7, Proposition I.3]. □

Theorem 49 (fixed height)**.** *Let $h \geq 2$. (a) There are integers $m_0 \geq 3$, $p \geq 1$, and $k_0, \ldots, k_{p-1}$ such that $f(h,m) = m + k_{m \bmod p}$ for every $m \geq m_0$. (b) The series $\sum_{m \geq 0} a_h(m) z^m$ is rational, with the conventions of Section 2, and the words of optimum blackouts form a regular language.*

Proof. Let $E(w) = |S(w)| - |w|$. Theorem 8 and $|S(v)| \le h|v|$ bound the excess of every factor of a valid word by $C_h = \max(q_h, 2h-2)$. So a counter for the excess of prefixes, kept in $[k - C_h, C_h]$, gives an automaton for the valid words $\mathcal{L}_{h,k}$ of excess k, and their lengths are eventually periodic. For $m \ge 3$, Lemma 16 and Theorem 8 make $f(h,m) - m$ the largest k from $h-1$ to q_h with a word of length m in $\mathcal{L}_{h,k}$, which gives (a). The optimum words are those of $\mathcal{L}_{h,k_r}$ of lengths $m \ge m_0$ with $m \equiv r \pmod p$, and finitely many shorter ones, so (b) follows as in Theorem 48. □

Guilhot, Little, and Parkinson [9, Theorem 1(3)] prove that the words of maximal weight in a regular language form a regular language for an additive weight bounded above. This does not give (b), since we maximize at each width separately. Longer proofs are in `jis/removed/sec08_height.tex` of the repository [20].

9 Computations and verification

The results that rest on a computation are labeled computer-assisted. Every other result in Sections 2 to 8 has a written proof, unless a remark says that its proof is sketched or omitted. Only Remark 35 of these is used later, in the design of a count. Every computed value and count comes from at least two programs written separately, which agree wherever both ran. Before most computations a prediction went into the ledger `kagey-problems/problem001/data/ledger.md`, and several were wrong, among them $f(5,5) = 9$, $f(4,8) = 13$, $f(5,8) = 13$, and $f(6,8) = 14$.

All programs are in Python 3.14.3. The ILP program `solver/engine.py` finds a smallest hitting set for the difference sets of Lemma 2 with CBC through PuLP [16]. It took 723 seconds on the 5×7 grid and 1,063 seconds on the 6×6 grid. The brute-force check `solver/brute.py` uses no lemma and shares no code with it. The verifier `verify_narrow.py` checks the lemmas and constructions of Sections 4 to 6 and the 5×5 grid in 21 seconds. The blackouts of $h + m$ points on the grids 5×6, 5×7, 6×6, 6×7, and 7×7 are stored in `jis/data/small_cells_sat_*.json`. The SAT solvers are CaDiCaL 1.5.3 and 1.9.5, Glucose 3, 4, and 4.2, MiniSat 2.2, MapleChrono, and Lingeling from pysat 1.9.dev15, and the algebra uses SymPy 1.14.

A SAT instance asks for a valid blackout of at least k points, so by the observation after Corollary 3 an unsatisfiable instance proves $f(h,m) < k$.

Encoding A has a variable for each point and a clause "not all black" for each inclusion-minimal symmetric difference of two rectangles, with a sequential counter for the size. Encoding B has a clause for each pair of rectangles that share a corner and an indicator for each hidden rectangle, at most one of them true, with a totalizer for the size. The third encoding, `cells_sat_check.py`, is of type A but written from scratch. The structured encoders of Section 7 add the column structure of Theorem 11 for one row partition and keep only the clauses that involve a tilted rectangle. The collar test boards of Section 7 are checked by three separately written enumerators, which list every rectangle of the board and compare the presentations. The column searches use no solver. They place columns from left to right and stop when two rectangles inside the placed columns show the same points, which is sound since every interval of columns of a valid blackout is valid. We did not produce DRAT proofs, and rely on the agreement of independent encodings on different solvers.

The scripts in `jis/scripts` form five groups, by the prefixes `three_row`, `four_row`, `five_row`, `six_row`, and `small_cells` or `cells`. The file `jis/scripts/README.md` lists the scripts behind each computer-assisted result and the longest run of each, at most 1,578 seconds. Table 2 collects the values, and the rectangle counts of the programs agree with A085582 and A289832. The Lean 4 file `Problem001.lean` in `kagey-problems/lean/KageyProblems` formalizes Section 2 and the Strip Theorem with no `sorry`. Code and data are in the folder `kagey-problems/problem001` of the repository [20].

grid	f	$a_h(m)$	by	grid	f	$a_h(m)$	by	grid	f	$a_h(m)$	by	grid	f	$a_h(m)$	by
2×2	4	1	HIB	3×5	7	892	IB	4×6	10	26	IB	5×10	16	350	SC
2×3	4	12	HIB	3×6	8	3,420	IB	4×7	11	292	IB	5×11	17	3,004	SC
2×4	5	32	HIB	3×7	9	12,704	IB	4×8	12	2,118	IMD	5×12	18	24,706	SCM
2×5	6	80	IB	3×8	10	45,864	IB	5×5	10	8	IB	6×6	12	–	IS
2×6	7	192	TI	3×9	11	161,992	IB	5×6	11	–	IS	6×7	13	–	S
2×7	8	448	TI	3×10	12	562,912	ITD	5×7	12	–	IS	6×8	15	–	S
3×3	5	53	IB	4×4	7	316	IB	5×8	14	4	SDC	6×15	24	–	S
3×4	6	224	IB	4×5	8	1,810	IB	5×9	15	44	SC	7×7	14	–	S

Table 2: Computed values for $h \le m$ and the methods that agree. H is by hand, T a theorem, I the ILP program (value only on 3×10, 4×8, 5×6, 5×7, and 6×6), B brute force, D every equality candidate, C a column search, S two SAT encodings, M a transfer counter, and a dash not computed.

10 Open questions

(1) *Five-row counts.* The series $\sum_{m\ge 0} a_5(m) z^m$ is rational by Theorem 49. Find it, and the growth rate of $a_5(m)$ (Remark 42).

(2) *Six rows below width* 15. By the stored witnesses `jis/data/small_cells_sat_6x9_b16.json` and `small_cells_sat_6x10_b17.json`, the bound $f(h,m) \ge h+m-1$, and Theorem 43, $f(6,9)$ is 16 or 17, $f(6,10)$ is 17 or 18, and $m+5 \le f(6,m) \le m+8$ for $11 \le m \le 14$. What are these values?

(3) *Seven and more rows.* Is the row bound $m + q_h$ attained at every large width for every $h \ge 7$? By Theorem 39, one valid $7 \times m$ blackout of $m+12$ points that passes the collar test would settle $h = 7$.

(4) *Squares.* For some $c > 0$ and all large n, $c\, n^{6/5}(1+\ln n)^{-2/5} \le f(n,n) = O(n^{4/3})$ by an alteration argument [2, Section 3.2] (proof omitted, see `jis/removed/appC_square.tex`) and by Neuwirth [17, Corollary 16]. What is the true order, and the least n with $f(n,n) > 2n$?

(5) *Proofs by hand.* Is there a proof without a computer that the 5×5 optimum is unique up to symmetry, or that the four-row generating function has a dominant pole of order five?

11 Acknowledgments

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A Three-row generating functions

We finish the proof of Theorem 25, with $D = LN = 1 - 2z - 4z^3 - 4z^4$. A *restricted run* is a string of singletons with no two letters b at distance two. Reversal and reflection carry the next lemma to BA, BC and CB.

Lemma 50 (forbidden windows in the classes containing B)**.** *Let S be a candidate in the ordered class AB or in the triple class. A window fails if and only if the table lists it for its class, where s is any singleton.*

class	*windows*	*forbidden*
AB, X	*singletons only*	*bab, bbb, bcb*
AB	*one double*	*bsA, Asb, bAb, bAc, cAb, aBa, aBb, bBa, cBc*
	two doubles	*bAB, ABa*
X	*one triple*	*bsX, Xsb, aXa, aXb, bXa, bXc, cXb, cXc*

Proof. The tests (a) to (e) are those of Lemma 20, and Lemma 21(b) gives the singleton windows. A window sAt behaves as in Lemma 22(a), except that test (e) also fails bAb, by $1 \in B_{i+1}$ and $B \in F$. In sBt and sXt column $i+1$ is black. The diamond is empty if $s = t = b$. If $s = b \neq t$ it shows $(i+2, 2)$ alone and fails by test (e) exactly when $\{2, t\} \in F$, and symmetrically for $t = b \neq s$. If s and t are outer it shows the horizontal pair and fails by test (c) exactly when $s = t$. In uAB the diamond fails by test (d) for $u = b$, since $(i+2, 3)$ and $(i+1, 2)$ are black, and shows three corners otherwise. In ABw it is empty for $w = b$ and otherwise fails by test (e) exactly for $w = a$. In Asb and Xsb it shows the vertical pair or one point of column $i+1$, and tests (b) and (e) fail it as $B \in F$. Every other window with a double or X at the left tip shows three corners or an oblique pair whose test (d) needs the white point $(i+1, 2)$, so it passes. Reversal gives the right tip. □

We count restricted runs by a transfer on the last two bits. An outer letter has the bit 0 and the weight 2, and b has the bit 1. Let h_{ij} count, by z^n, the continuations of length $n \geq 0$ of a restricted run ending in the bits i, j. Since b is forbidden after $1, j$, $h_{00} = 1 + z(2h_{00} + h_{01})$, $h_{01} = 1 + z(2h_{10} + h_{11})$, $h_{10} = 1 + 2zh_{00}$ and $h_{11} = 1 + 2zh_{10}$, and eliminating gives $Dh_{00} = 1 + z + 3z^2 + 2z^3$. Read runs $w_1 \cdots w_n$ with $n \geq 1$ outward from a double. Let g_0 count those with w_1 a prescribed outer letter and g_1 those with $w_1 = b$. Let u and v count the same runs with w_2, if present, outer. A fictitious outer letter before w_1 restricts nothing, so $g_0 = zh_{00}$, $g_1 = zh_{01}$, $u = z + 2z^2h_{00} = zh_{10}$ and $v = z + 2z^2h_{10} = zh_{11}$. Hence

$$Dg_0 = z(1+z+3z^2+2z^3), \quad Dg_1 = z(1+z), \quad Du = z(1+2z^2+2z^3), \quad Dv = z. \tag{12}$$

Proposition 51 (the triple class)**.** *In the notation of* (12)*,* $G_X(z) = z\big(1 + 4u + 2v + 2u^2 + v^2\big)$ *has a denominator dividing* D^2*.*

Proof. In UXW the runs are restricted, and bsX and Xsb make the second letters from X outer (Lemma 50). So one nonempty run gives $2u + v$ on

either side. With two, the window at X allows only the neighbor pairs (a,c), (b,b) and (c,a), which give $2u^2+v^2$. $\square$

A word of AB is $UAVBW$ with restricted runs U, V and W. By Lemma 50, a nonempty U read from A has an outer second letter and is counted by u or v, and a nonempty W read from B is counted by g_0 or g_1. Let ℓ_t and r_t count the allowed U and W, the empty one included, when a singleton t stands right of A or left of B. The windows with the double in the middle give

$$(\ell_a,\ell_b,\ell_c)=(1{+}2u{+}v,1{+}u,1{+}2u),\ (r_a,r_b,r_c)=(1{+}g_0,1{+}g_0{+}g_1,1{+}g_0{+}g_1). \tag{13}$$

For $s\in\{00,10\}$ and a bit β let $J_{s\beta}$ count, as for h, the continuations from s whose last letter has the bit β. Solving as for h gives

$$J=\frac{1}{D}\begin{pmatrix}1+2z^2+2z^3 & z+z^2\\ 1 & 2z^2+2z^3\end{pmatrix}. \tag{14}$$

Proposition 52 (the class $\{A,B\}$, polynomial form computer-assisted)**.** *In the notation of* (12) *to* (14)*,*

$$\begin{aligned}G_{AB}(z)=z^2(1+2u)(1+g_0+g_1)+z^3(\ell_a r_a+\ell_b r_b+\ell_c r_c)\\ +z^4\,(\ell_a+\ell_c,\ \ell_b)\,J\,(r_a+r_c,\ 2r_b)^T.\end{aligned} \tag{15}$$

Moreover $G_{AB}=P_B/D^3$ *with*

$$P_B(z)=z^2(1+z)(1+z^2-3z^3+4z^6+4z^7). \tag{16}$$

Proof. The runs cover the windows inside them, and we split by the length k of V. For $k=0$ the windows uAB and ABw require $u\neq b$ and $w\neq a$, which gives the first term. For $k=1$ the window AtB passes, and uAt and tBw give ℓ_t and r_t. For $k\geq 2$ the window Av_1v_2 makes v_2 outer, $v_{k-1}v_kB$ passes, and the weight is $\ell_{v_1}r_{v_k}$. After v_2 the bits are 00 or 10 as v_1 is outer or b, and the rest is a continuation in which v_2 keeps its factor 2 for $k\geq 3$. Summing over the last letter gives $J_{s0}(r_a+r_c)+J_{s1}\,2r_b$, and $k=2$ is the constant term of J.

The script `three_row_closed_form.py` finds (16) in SymPy. Its first coefficients 7, 31, 120, 449 and 1622 match `three_row_definition_check.json`. $\square$

Theorem 53 (the exact denominator, computer-assisted)**.** *The reduced denominator of* $\sum_{m\ge 3} a_3(m)z^m$ *is* $Q = R^3L^3N^3$*, with five triple poles.*

Proof. By Theorem 25(a) the denominator divides $R^3D^3 = Q$. The factors R, L and N are coprime, since $L - N = -2z - 4z^2$ vanishes only at 0 and $-\frac{1}{2}$, where $N \neq 0$, and L and N are positive at $\frac{1}{3}$. The numerator in (2) is $\frac{4}{81}$ at $z = \frac{1}{3}$, where G_{AB} and G_X are analytic. Division gives $P_B \bmod L = \frac{z}{2}$ and $P_B \bmod N = \frac{1}{4} - \frac{z}{2}$, so P_B vanishes at no root of L or N. At these roots G_{AB} has a pole of order three, G_X one of order at most two, and G_{AC} none. □

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